A Nonlinear Systems-Theoretic Framework for Transient Hyper-Regenerative States in Adult Mammals: Evolutionary Constraints, Stability Analysis, and Oncogenic Boundary Conditions "OBRS"

Abstract

Adult mammals display limited regenerative capacity relative to regenerative vertebrates such as the Axolotl and *Danio rerio*. Comparative molecular analyses reveal conservation of canonical developmental pathways across vertebrates, suggesting that regenerative restriction in mammals is predominantly regulatory rather than genomic.

We propose a nonlinear dynamical systems framework describing a Transient Hyper-Regenerative State (THRS) as a metastable attractor in adult mammalian tissue. The model integrates proliferative plasticity, epigenetic accessibility, fibrotic constraint, morphogenetic coherence, oncogenic load, and surveillance amplification within a coupled differential system.

We formalize the Regeneration–Oncogenesis Tradeoff Hypothesis (ROTH) as an evolutionary stability optimization between plasticity and genomic integrity. Stability analysis reveals that hyper-regeneration is dynamically sustainable only if oncogenic surveillance scales superlinearly relative to proliferative activation. Otherwise, the system undergoes bifurcation into a malignant attractor regime.

This framework unifies regenerative biology, evolutionary theory, and tumor suppression dynamics under a common systems-theoretic architecture.

1 Conceptual Foundation

1.1 Conservation vs Suppression

Vertebrate genomes retain conserved morphogenetic modules:

- Wnt
- FGF
- Hedgehog
- TGF-β
- Notch

The disparity between regenerative and non-regenerative species likely emerges from network-level regulatory architecture, not absence of structural genes.

Thus, regeneration must be modeled as:

- A state-dependent attractor configuration of a high-dimensional regulatory network.

2 State Variables and Normalization

We define continuous, normalized variables in $[0, 1]$:

Symbol	Meaning
P	Proliferative plasticity
E	Epigenetic accessibility
F	Fibrotic constraint
M	Morphogenetic pattern coherence
O	Oncogenic instability load
S	Surveillance intensity
I	Injury/inflammatory stimulus

3 Regenerative Output Functional

We define regeneration as multiplicative synergy:

$$R(t) = P^{\alpha} E^{\beta} M^{\gamma} (1 - F)^{\delta} \tag{1}$$

With positive exponents ensuring nonlinear sensitivity.

This captures:

- Regeneration collapses if any critical axis approaches zero.
- Fibrosis acts as inhibitory field.

4 Core Dynamical System

4.1 Proliferative Plasticity

$$\frac{dP}{dt} = A_1 E(1 - F) - a_2 OP - a_3 P \tag{2}$$

Interpretation: Induced by epigenetic permissiveness and low fibrosis; suppressed by oncogenic stress; naturally decays toward baseline.

4.2 Epigenetic Accessibility

$$\frac{dE}{dt} = B_1 P - b_2 OE - b_3 E \tag{3}$$

Interpretation: Coupled positive feedback with P; negative feedback via oncogenic load.

4.3 Fibrotic Constraint

$$\frac{dF}{dt} = C_1 I - c_2 P - c_3 F \tag{4}$$

Interpretation: Injury drives fibrosis; regenerative plasticity suppresses fibrosis; matrix remodeling decays F naturally.

4.4 Oncogenic Instability

$$\frac{dO}{dt} = D_1 PE - D_2 SO \tag{5}$$

Interpretation: Mutation pressure proportional to proliferative epigenetic activity; surveillance removes unstable clones.

4.5 Surveillance Amplification

$$\frac{dS}{dt} = E_1 P^k - E_2 S \tag{6}$$

Critical condition: $k > 1$. Superlinear scaling required for stability.

4.6 Morphogenetic Coherence

$$\frac{dM}{dt} = F_1 P - f_2 O - f_3 M \tag{7}$$

Interpretation: Pattern formation requires proliferation but degrades under genomic instability.

5 Equilibrium Analysis

Set derivatives to zero. From oncogenic equilibrium:

$$O^* = \frac{D_1 P^* E^*}{D_2 S^*} \tag{8}$$

Substitute into plasticity equation:

$$0 = A_1 E^*(1 - F^*) - a_2 \left(\frac{D_1 P^* E^*}{D_2 S^*} \right) P^* - a_3 P^* \tag{9}$$

Rearranging gives:

$$P^* = \frac{a_1 E^*(1 - F^*)}{a_3 + \frac{a_2 D_1 E^*}{D_2 S^*}} \tag{10}$$

This shows explicitly:

- Regeneration increases with surveillance strength.

6 Stability Conditions

Let $X = [P, E, F, O, S, M]^T$. The Jacobian matrix J governs local stability. System stable if:

$$\mathrm{Re}(\lambda_i(J)) < 0 \tag{11}$$

Critical instability arises when:

$$\frac{\partial}{\partial P}\left(\frac{dO}{dt}\right) > \frac{\partial}{\partial S}\left(\frac{dO}{dt}\right)\frac{\partial S}{\partial P} \tag{12}$$

Substituting:

$$D_1 E > D_2 O \cdot E_1 k P^{k-1} \tag{13}$$

If surveillance responsiveness insufficient $\rightarrow$ eigenvalue crosses zero $\rightarrow$ bifurcation into malignant regime.

7 Bifurcation Structure

Three attractors emerge:

1. **Fibrotic Stable State**: Low P, Low E, Low O, Moderate F.
2. **Hyper-Regenerative State (Metastable)**: High P, High E, Low F, Controlled O, High S.
3. **Malignant Attractor**: High P, High E, High O, Surveillance collapse.

Transition from $2 \rightarrow 3$ occurs if $k \rightarrow 1$ or $E_1 \rightarrow 0$.

8 Evolutionary Optimization

We define fitness functional:

$$\Phi = \int_0^L (W_1 R(t) - W_2 O(t))\, dt \tag{14}$$

Long-lived species maximize: $\frac{\partial \Phi}{\partial P} \to 0$.

Thus optimal evolutionary solution may favor $P_{\text{low}}, S_{\text{moderate}}$ over $P_{\text{high}}, S_{\text{extreme}}$. Energetic cost of sustained high surveillance likely scales as:

$$\text{Cost}_S \propto S^m, \quad m > 1 \tag{15}$$

9 Neural Extension

Introduce circuit stability C:

$$\frac{dC}{dt} = G_1 M P - g_2 O - g_3 P^2 \tag{16}$$

Excess plasticity degrades coherence quadratically. Thus whole-brain THRS unstable. Only spatially localized micro-THRS viable.

10 Aging Parameter Extension

Let $A(t) = \int_0^t O(\tau) d\tau$ be the cumulative genomic damage. Lifespan decreases as:

$$L \propto \frac{1}{A(t)} \tag{17}$$

Sustained hyper-regeneration accelerates aging unless $\frac{dO}{dt} \approx 0$.

11 Theoretical Conclusions

1. Mammalian regenerative suppression likely represents an evolutionarily stable attractor minimizing oncogenic bifurcation risk.
2. Hyper-regeneration is mathematically feasible only as a transient, spatially bounded metastable regime.
3. Surveillance amplification must scale superlinearly with proliferative activation.
4. Whole-organism hyper-regeneration is dynamically unstable.
5. Neural hyper-regeneration is particularly constrained by circuit entropy.